

FUNDAMENTALS OF COSTING AND PRICING

DAY 1



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LESSON ONE

COSTING TERMINOLOGY

TYPES OF COSTS

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CONTRIBUTION

LESSON ONE:

COSTING TERMINOLOGY

- Cost is a measurement, in monetary terms, of the amount of resources used for the purpose of production of goods or rendering services.
- Cost is the amount of actual or notional expenditure relating to a product, service, process or activity.
- Cost is often used as a generic term to describe various types of costs.
- Costing is the technique and process of ascertaining costs.

Costing is the “science” of ensuring that all inputs to the delivery of a product or service are covered.

Direct costs

All transport, labour and materials **directly used** in the manufacture of your product or the delivery of your service. Generally, if you are not making stock, producing an order or delivering a service, you are not incurring direct costs. These costs tend to vary month to month depending on your volume of work and are sometimes called variable costs.

Overheads

These are the running costs that one incurs whether you are making products, providing services or not – hence them also sometimes being called fixed costs. This includes things such as the cost of labour not directly involved in production or service delivery but needed to run the business (such as administrative staff, sales & marketing staff, cleaners etc), as well as all recurring and ad hoc running costs including electricity, internet, insurance, rent, phones, legal services, stationery, depreciation etc.

Cost of Goods Sold

Cost of goods sold (COGS) are the direct costs attributable to the production of the goods sold by a company. This amount includes the hard costs of the materials and direct labour used to produce the goods or deliver the service, or all the direct costs associated with purchasing stock for resale as in a retail environment. It excludes indirect expenses (overheads). COGS appears on the income statement and can be deducted from revenue to calculate a company’s gross margin (GP).

In manufacturing we call these Cost of Goods Sold, in retail it is usually referred to as “Cost of Sales.” This is essentially the same thing as Direct Costs, except sometimes COGS will include an apportionment for factory overheads, or storage of stock items.

Margins

Profit margin indicates the profitability of a product, service, or business. It’s expressed as a percentage; the higher the number, the more profitable the business.

Gross Profit Margin, or gross margin, is the difference between total sales and the direct cost/cost of sales of those goods. For example: If total sales equals R1,000 and cost of sales equals R300, then the gross margin equals R700 (or 70%).

Net Profit margin refers to the percentage of **revenue** remaining after all costs, depreciation, interest, taxes, and other expenses have been deducted. The formula is: $(\text{Total Sales} - \text{Total Expenses}) / \text{Total Sales} \times 100 = \text{Net Profit Margin}$ as a percentage.

Break-Even Point

A level where the amount of sales attained is met with the same amount of costs, resulting in zero profit or loss. A formula (which is in your addenda) can be used to determine the break-even point for a business.

Depreciation

Depreciation is a concept used for tax and accounting purposes that describes the method a company uses to account for the declining value of its assets. The reasoning? An asset acquired this year is unlikely to be worth as much in five years because it often will have worn down, been depleted, or become obsolete.

Depreciation is an accounting convention that allows a company to write off an asset's value over time, but it is considered a non-cash transaction.

As a benchmark, most general assets are depreciated at a rate of around 20 – 25% per year.



TYPES OF COSTS

DIRECT (OR VARIABLE) COSTS

- These are costs that are only incurred **if you work on a particular product, job or order**, and include all material costs (including things such as venue hire, accommodation or catering if you provide a service), direct labour and any other materials (eg boxes, packaging, pallets, printouts) required to complete that particular product, job or order.
- You can also include transport as a direct cost, provided it relates specifically to that job or contract eg physical delivery. General transport would be listed as an overhead (see below).
- In some cases, you may find it appropriate and useful to allocate items such as electricity usage as a direct cost, if these are able to be measured accurately and traced to a specific product/project – such as when using a kiln or a commercial oven that can be monitored.

OVERHEADS (FIXED COSTS)

- Overheads/fixed costs are costs which are incurred whether you are producing orders, selling products, providing services or not. They may fluctuate slightly month to month depending on work volumes, but in general these are stable, recurring costs.
- Overheads are frequently left out of the costing process or miscalculated, leading to an incorrect final costing and potential losses as a result. They need to be accounted for.
- Overheads include costs like:
 - o Telephone and internet accounts
 - o Advertising and promotional costs
 - o Electricity and water
 - o Rent/bond/rates
 - o Accounting and other business services
 - o Fixed salaries and wages (marketing, finances, admin & accounts etc)
 - o Vehicle and machinery maintenance and depreciation – usually calculated as a percentage of the total value of the item/s
 - o Insurance – theft, fire, key man, loss of earnings
 - o Consumables (computer software, small hand tools, machinery repairs and renewals)
 - o General transport and fuel costs
 - o Staff training and development
 - o Uniforms & PPE wear
 - o Etc

NOTE: Some businesses divide their overheads up into Manufacturing/Factory Overheads and Administrative Overheads. For this workshop, we will combine these two into one figure.

OVERHEAD ALLOWANCE OR OVERHEAD CONTRIBUTION

Because in most instances it is nearly impossible to apportion exact overhead usage to a specific product or service, a viable option is to use an **Overhead Contribution** to apportion overhead costs. It is called an overhead *contribution* because every product sold or service delivered by your organisation needs to give something back/contribute towards covering the running costs of the business, by being included in the costing calculation process.

So how do we record it?

The most simple and accurate way of recording an overhead contribution, is to express it as a percentage. This percentage is added to your **direct costs** (direct labour & materials) to give you a TOTAL COST to deliver that product or service - per product made, per learner trained, per shipment delivered, per pump serviced, per contract executed etc. The formula looks like this below:

$$\text{Direct Costs (Labour \& Materials) + Fixed Costs (Overhead Contribution as a \%)} \\ = \text{Total Cost of Production/Delivery of Service}$$

How much should you add?

Every business has different overhead and cost structures, so there is no 'one-size-fits-all' answer. In a service-based business, your main costs are usually staff related, so the figure will depend on how large and expensive your team is. In a production environment, a typical overhead figure would range anywhere from 25% to 70%, and in agriculture this could be even higher due to the need to recoup running costs for expensive machinery and equipment, depreciation, maintenance and other general running costs.

The figure is generally slightly less for the services sector, which although quite heavy in terms of staff costs as indicated above, usually has less equipment & machinery costs, less need for a large production or workspace, lower electricity usage etc than a large-scale manufacturer with a big factory or a farmer, for example.

At the end of the day, your overhead contribution % all depends on how high your overheads are that need to be recovered every month/quarter/year.

For the purpose of this workshop, we will use an overhead contribution of 25% - this is a bit lower than average but will serve our purpose as an example in our calculations.

As the formula above shows, this % figure is **added to your direct costs** to get to the **total cost** of the product or service being costed. In this way, every product you sell, or every service you provide, gives back (contributes) a portion towards the running costs of your organisation.

Note: *It is quite straightforward – and very important - to work out **exactly** what the percentage for O/H contribution should be in your specific business – see the formula below:*

Calculating Your Overhead Contribution Percentage

Every business has a different cost recovery strategy, but there is a useful formula you can apply should you wish to work out the exact overhead contribution percentage for your business in its current form:

STEP ONE:

List all **Direct Costs and Overheads** for the year in two separate columns, and then add each of them up to get a total for Direct Costs for the year and Overheads for the year. (Note – this can be done for a Quarter if you do not have all info at hand, but a full year is better if possible).

DIRECT COSTS

Raw Materials
Packaging
Production Labour Costs
Delivery Costs

OVERHEADS

Telephone & Internet
Advertising
Legal
Rent
Electricity and Water
Accounting Services
Fixed Salaries/Wages
Consumables
Depreciation @ 25% etc

STEP TWO:

Divide the Overhead total by the Direct Costs total, and multiply by 100.

Let's assume that the Direct Cost figure is R600 000 and the Overhead column adds up to R450 000 (for example). Simply divide the Overheads by Direct Cost i.e. $450\,000 / 600\,000$ and multiply by 100. In this example you get a figure of 75% and this tells you that you have to add 75% to the **direct cost** of every item to work out the true cost of making it.

For example, let's say you make a product at a Direct Cost of R10.00. You would then add the 75% (R7.50) to this direct cost figure and establish it really cost you R17.50 to make when you take into account your overheads. If you sell it at less than R17.50 you will soon go bust - Everything above R17.50 you sell it for is your profit.

Allocation of Overheads - Remember to correctly allocate overheads to the **right cost centre** – this is especially important for businesses with varied income generating activities.

For example, if you make bricks and also provide training in construction services, when costing either cost centre you should only allocate the **relevant overheads** incurred through the development, production, marketing and selling of that **particular activity** to your costs, not the whole overhead obligations of the entire organisation. Otherwise, you will skew the figures and/or end up subsidizing one or other of the business units without necessarily realising it.

It is important to cost in the correct amount for overheads because:

- Failure to measure and cost in your actual overheads will usually result in your product or service being under-priced
- Not costing in enough for overheads will almost certainly result in minimal profits and potentially cause cash flow issues
- If you know your overheads and direct costs, you can accurately work out exact profits per product, per contract, per harvest or per service delivered. This is very empowering information for an entrepreneur.

To remind you, this leaves us with:

$$\text{Total Cost of Production/Delivery of Service} = \text{Direct Costs (Labour/Materials/Packaging)} + \text{Fixed Costs (Overhead Contribution as a \%)}$$

Example:

	Total Cost
Total Direct Labour Costs	R100
Total Direct Material Costs	R120
Total Direct Costs	R220
Plus Overhead Contribution @ 25%	$R220 \times 25\% = \text{R55}$
Total Cost of Production	R275

In the above example we have costed in R55 to this product to give back towards the running costs of the company.